

Spec-Driven Development

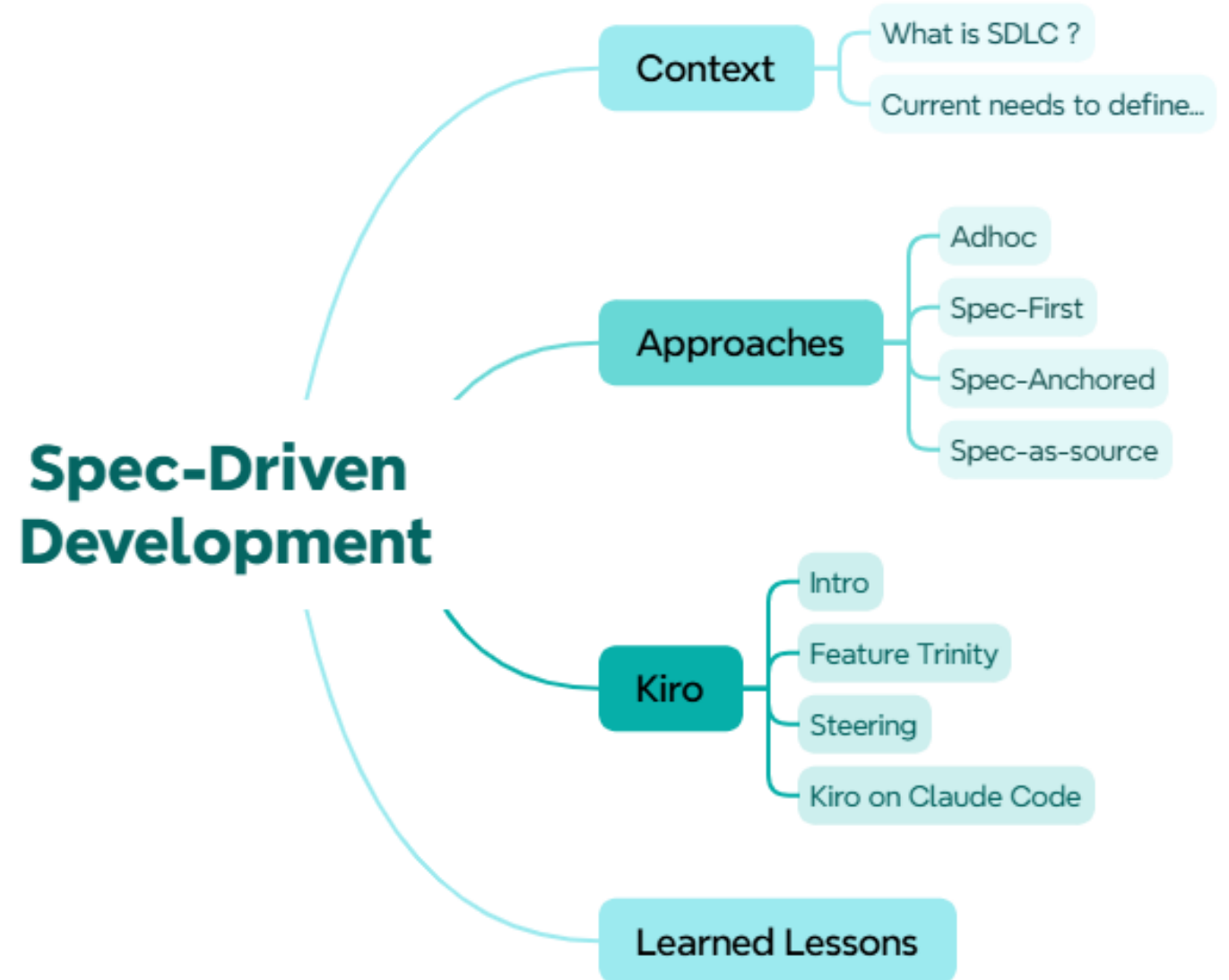
Arquitectura de Software II



Universidad de
La Sabana

FACULTAD DE INGENIERÍA

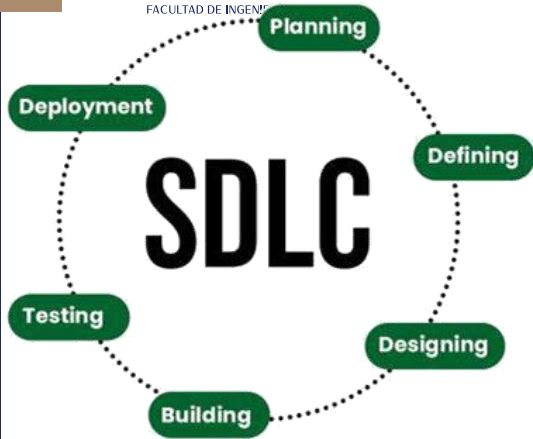
Agenda



My interest is to choose a simple framework to reduce number of iterations (time and token consumption).

Out-of-Scope: Agentic Workflow: SPD+SDD+Harness

What is SDLC ?



Vibe Coding

Reactive,
prompt-heavy
and fragile

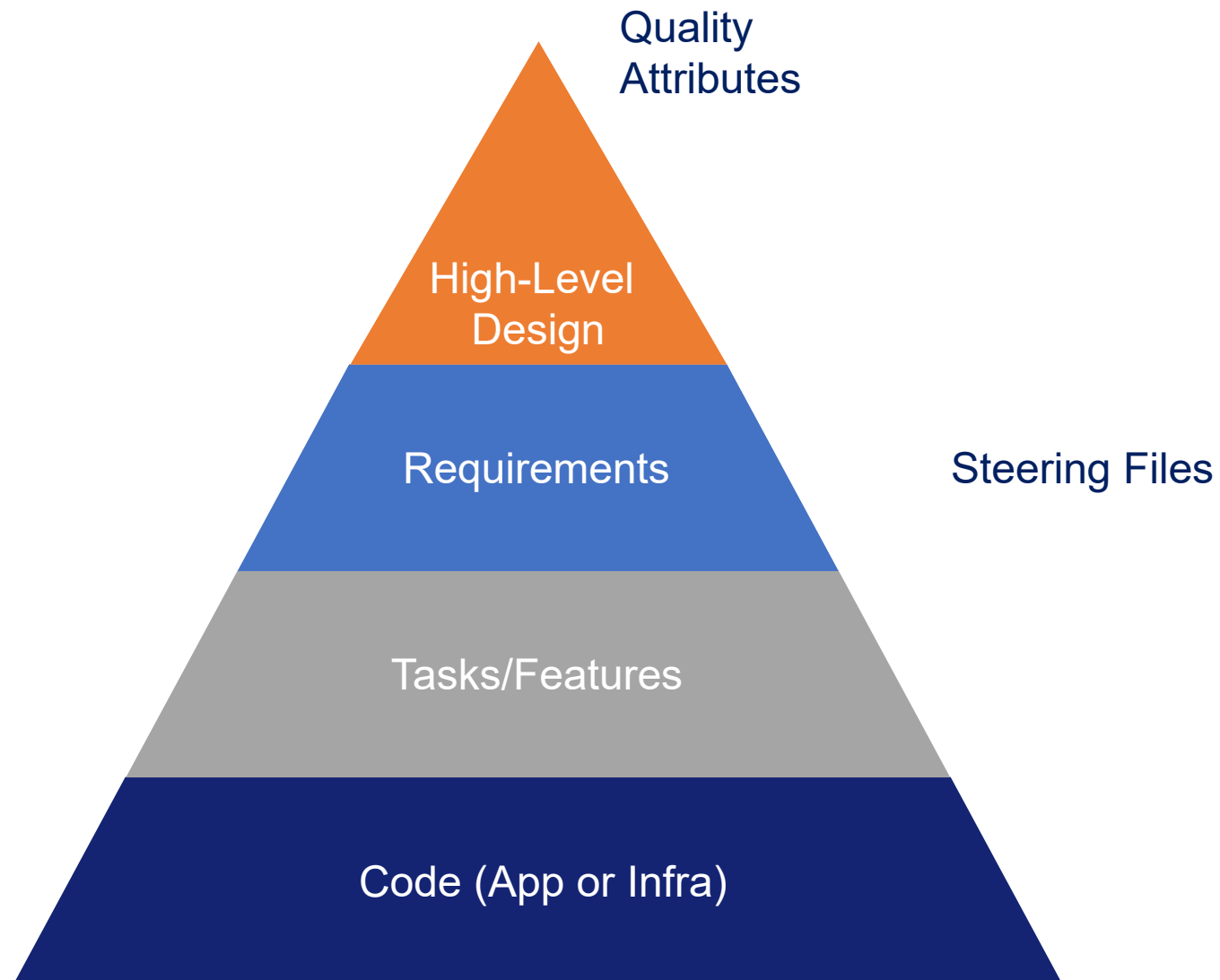
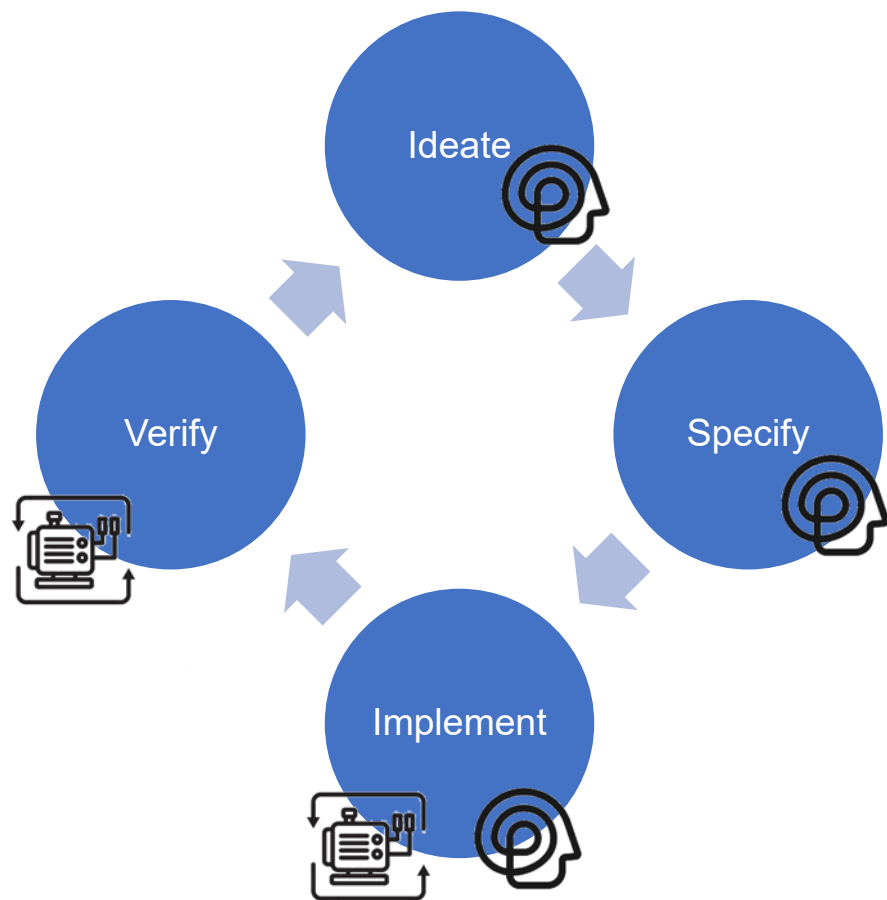
SDD

Documented and
verifiable

Feature	Waterfall	Agile	Vibe Coding (Chat AI)	Spec-Drive Development (SDD)
Role of Doc	Rigid Contract	Minimal Guidance	N/A	Executable Blueprint (Living Doc)
Context Management	Contract	Human Memory	Token Window	Local Index & Spec Files
Source of Truth	Front Doc	Working Software		Living Specification
Repeatability	Components/Libraries	Manual	Hit or Miss	Deterministic Plans
Code Quality	Human PR Review	Human PR Review	Review?	Spec Validated

Needs to define...

Objective: Define an organized method to manage the migration/reorg of workloads.



Approaches for SDD

Evolution of BDD and TDD

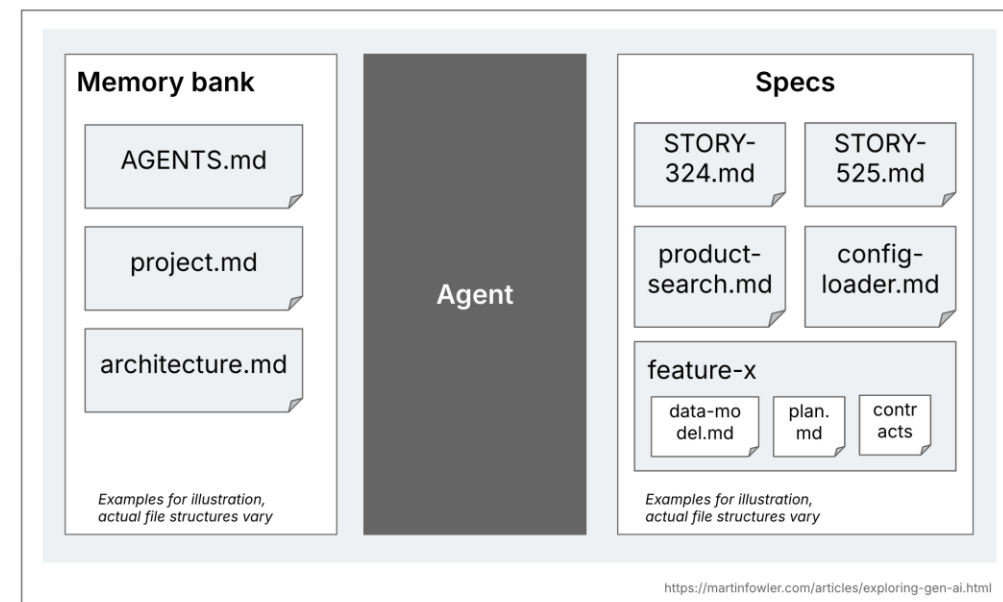
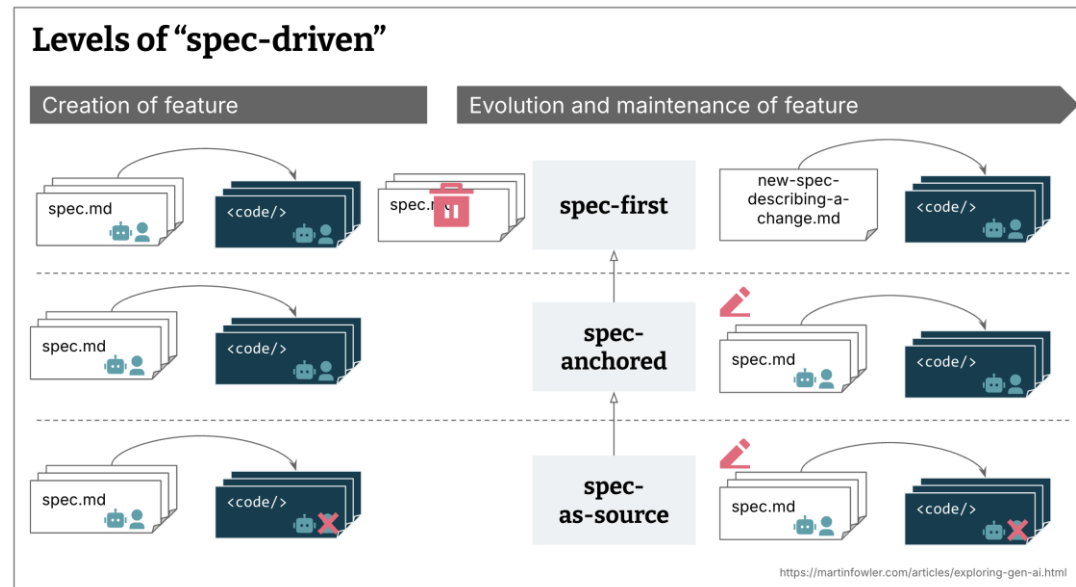
3 approaches:

- Spec-First
- Spec-Anchored
- Spec-as-source.

Differences due to modifications on initial specifications and code.

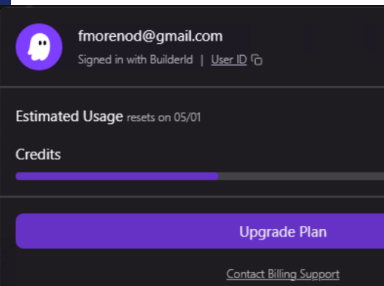
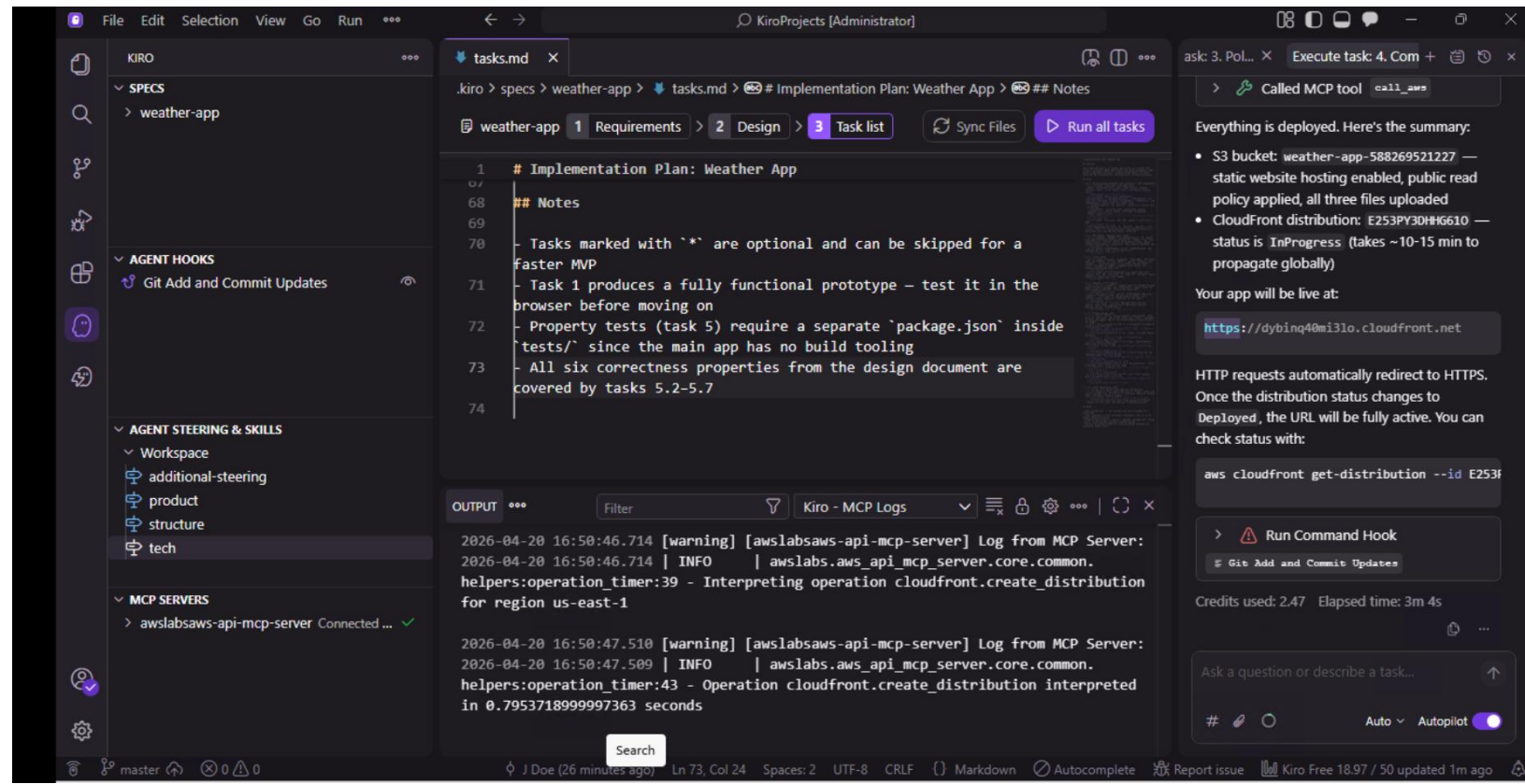
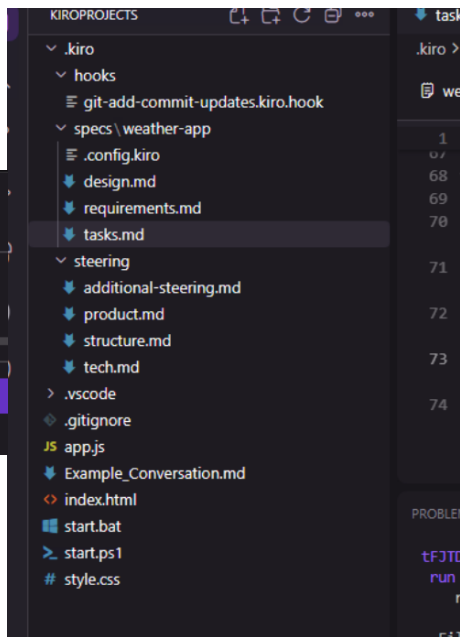
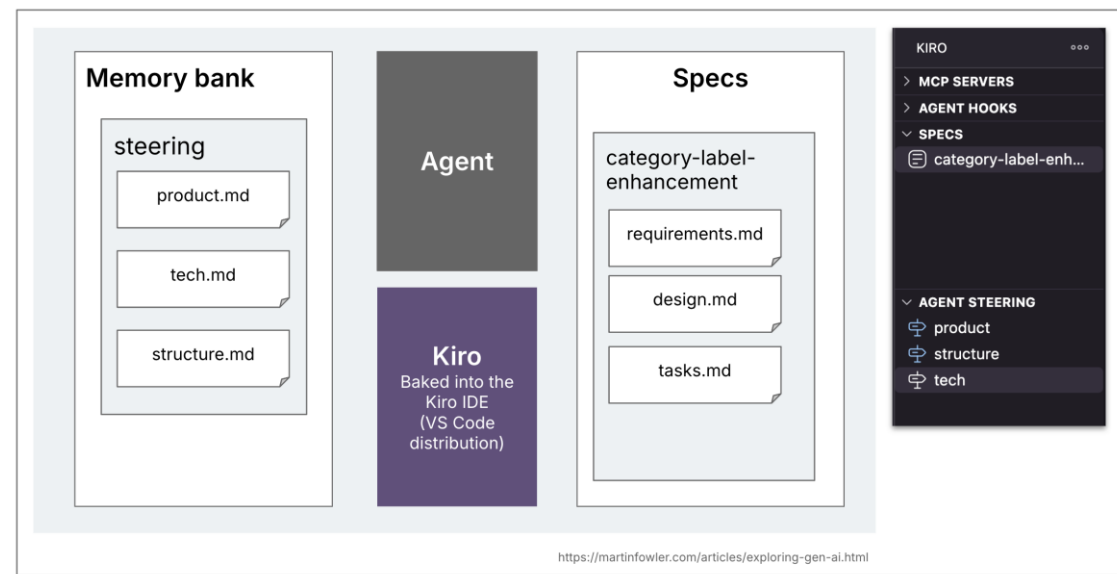
Structured similar: Memory bank, specs, features/tasks

There are another SDD frameworks, well-mature such as GitHub [SpecKit](#), or [OpenSpec](#); or methodologies such as BMAD. For simplicity, I use Kiro.

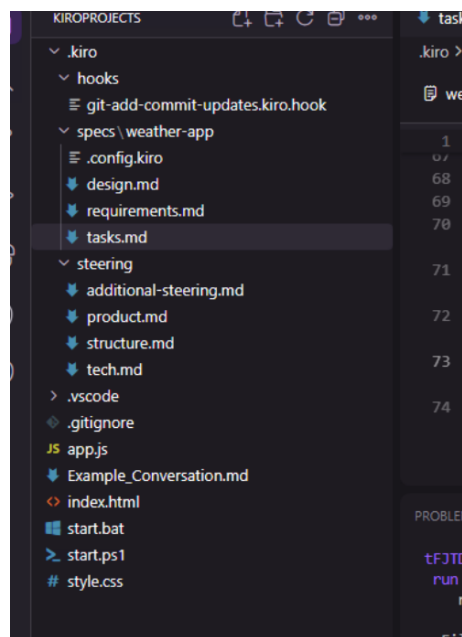




- AI IDE with focus on Spec-Driven Dev, for features but support bug fixing.
- AWS managed and charge LLM.
- AWS Support and Training
- Simple and intuitive to learn



Kiro: Feature Trinity/Three-Phased Workflow



Initial Prompt

Requirements

- Functional Requirements. Captures user intentions and its constraints. Defined what needs to do it in EARS syntax.

Design

- Schema abouts how to do it. State machines and relationships.

Tasks

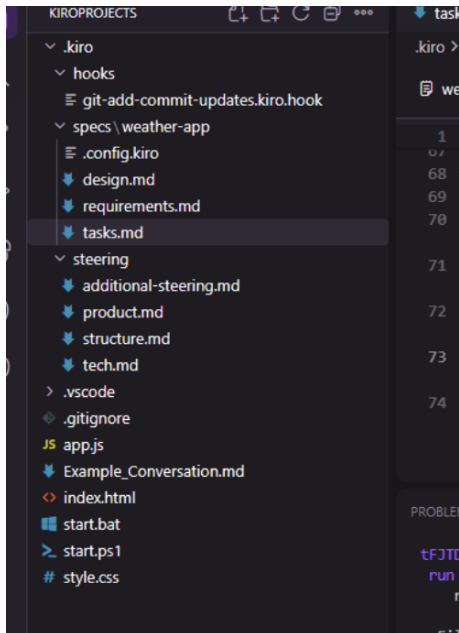
- Executable plan. Atomic sequence plan that agent has to follow.

Kiro: Steering

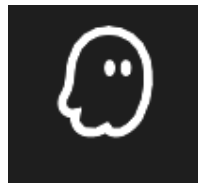


Define general rules that aren't link to the feature directly.
There are examples for government and tech definitions, such as:

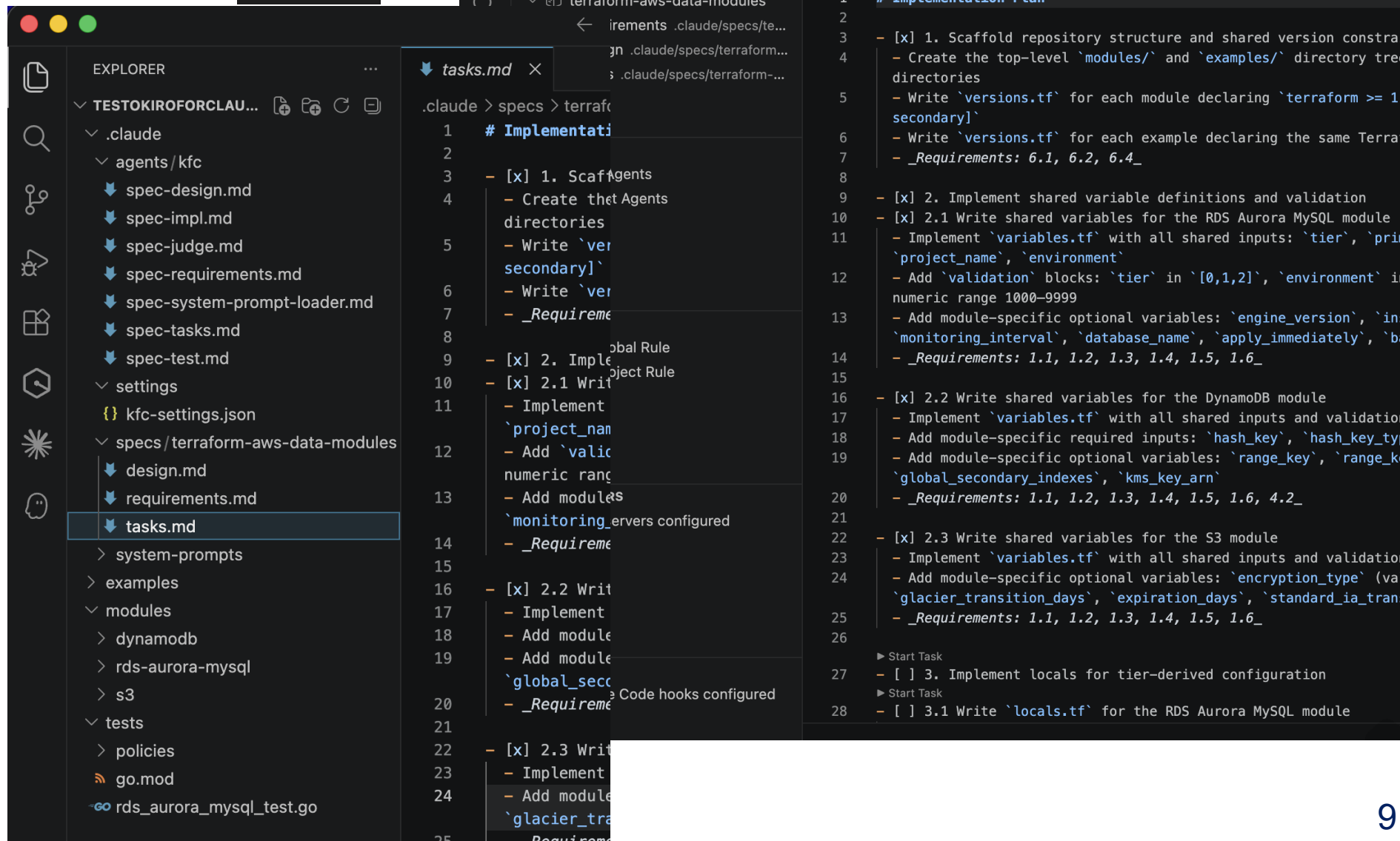
- Tech Stack: Frontend with React and Tailwind CSS. Backend with Python 3, FastAPI. DB with PostgreSQL and Prisma as ORM
- Patterns: Using Domain-Driven Development. All code has to have a header documentation about purpose of the file and methods.
- Naming definition: Use PascalCase for components names and camelCase for file names.



Kiro on Claude Code



- VSCode Extension
- Need a Claude Code API Keys
- Bypass Security to act as agent
- Create more under-the-hood files (.claude/agents)

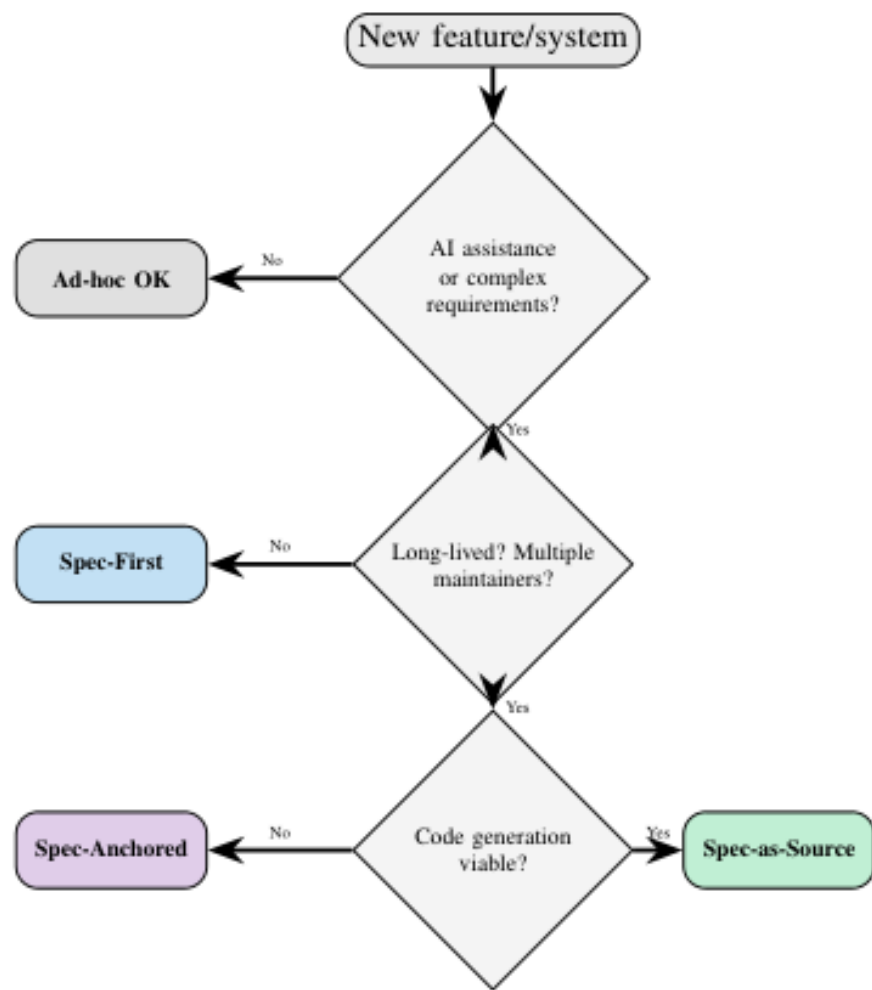


Out of Scope

My recommendations are:

- **Additional loops:** AI Harnessing (env), Context Eng(-as-a-Code; the product is the spec), Eval-Driven Design.
- **Branching strategy:** Use gitflow for complex projects, otherwise use trunk-based strategy for simplicity.
- **Context Window:** Depends on scale (member number) and distribution of the project, it can be a problem. Try to reduce limit information on the trinity or use limited information, i.e. [LLM Wiki](#) and [OpenViking](#)
- **Deployment:** Use several type of testing on early steps -focus on integration-, to avoid rework on CI/CD pipelines.
- **Human on the loop:** Consequences of avoiding this step: Reputational damage, dismissal, legal issues, bankrupt, or layoffs.

Learned Lessons



- Follow a method to scale on a team and choose the right approach based on team' maturity and maintainability.
- Need of an orchestrator (stages and transversal policies) between projects (Pipeline).
- Difference between steering and structured files (design, requirements, tasks)
- Allow optional tasks for testing (Validation tasks are longer) or using external tools.
- Modifications has to be saved on documents or on steering to have single source of truth, otherwise you can save interactions on Export Conversations.
- Cost is based on the complexity and tier.

References

About Spec-Driven Dev:
[arXiv \(Cornell\)](#) ,
[MartinFowler](#) and
[DeepLearning](#)

About Kiro:
SkillBuilder Courses [[1](#), [2](#)]

About Claude Code
Anthropic Courses [[A](#), [B](#)]

Spec-Driven Development: From Code to Contract in the Age of AI Coding Assistants

Deepak Babu Piskala
Seattle, USA
Technical Report

Abstract—The rise of AI coding assistants has reignited interest in an old idea: what if specifications—not code—were the primary artifact of software development? Spec-driven development (SDD) inverts the traditional workflow by treating specifications as the source of truth and code as a generated or verified secondary artifact. This paper provides practitioners with a comprehensive guide to SDD, covering its principles, advocated for by [15], is simple: poor at min Consider

Spec-Driven Development



[martinfowler.com](#)

Refactoring Agile Architecture About Thoughtworks

Understanding Spec-Driven-Development: Kiro, spec-kit, and Tessel



This article is part of "Exploring Gen AI". A series capturing Thoughtworks technologists' explorations of using gen ai technology for software development.

15 October 2025

Birgitta Böckeler



Birgitta is a Distinguished Engineer and AI-assisted delivery expert at Thoughtworks. She has over 20 years of experience as a software developer, architect and technical leader.



Structured Approach to AI coding with Spec-Driven Development on Kiro

Lab Information

- Not started
- 1 hour 30 minutes
- Available languages

Feedback

What's New

AWS Learning Assistant is here!

Use our AI-powered learning assistant to guide you through AWS code as you work through the lab.

Structure

Spec-Driven Development with Kiro

ANTHROPIC

Building with the Claude API

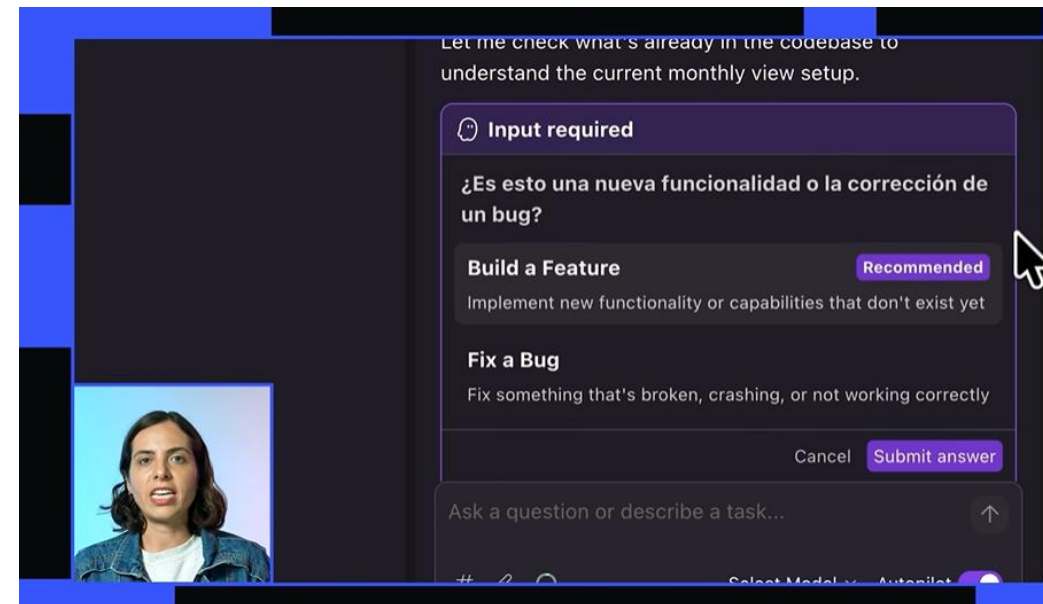
Demo

Build a cross-platform Currency Converter application using HTML, CSS (Tailwind via CDN), and vanilla JavaScript.

Key Requirements:

1. Allow the user to convert between USD, EUR, GBP, COP, and JPY.
2. Fetch live exchange rates from the free API:
<https://open.er-api.com/v6/latest/USD>
3. Include offline fallback rates if the API fails or is unreachable.
4. Clean, responsive UI with light/dark theme support according to system preferences.
5. Simple calculation history table showing the last 5 conversions.

Please generate a complete specification using the Spec-Driven Development workflow (Requirements -> Design -> Tasks).



	KIRO	CURSOR	BOLT/LOVABLE
APPROACH	SPEC FIRST, THEN CODE	CODE WITH AI ASSISTANCE	PROMPT TO FULL APP
SPEED	SLOWER START, FEWER REWRITES	FAST ITERATION	FASTEST TO PROTOTYPE
BEST FOR	PLANNED PROJECTS	DAILY DEVELOPMENT	QUICK VALIDATION
PRICE	\$19/MO PRO	\$20/MO PRO	\$20-25/MO
LEARNING CURVE	MODERATE	LOW IF YOU KNOW VS CODE	VERY LOW



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